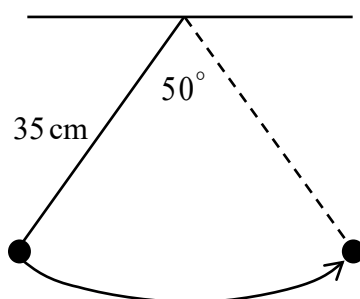


Question 1

2 marks 101

A pendulum that is 35 cm long swings through an angle of 50° . Determine the length of the arc through which the pendulum swings.



Question 2

4 marks 102

Solve algebraically, where $0 \leq \theta \leq 2\pi$.

$$2\cos^2\theta = \sin^2\theta - 2\cos\theta$$

Question 3

2 marks 103

Determine the number of arrangements of the letters in the word ATTENTION which begin with the letter A.

Question 4

3 marks 104

Solve for x , algebraically.

$$e^{2x+1} = 5^x$$

Question 5

2 marks 105

There are 10 teachers and 17 students who would like to attend a field trip.

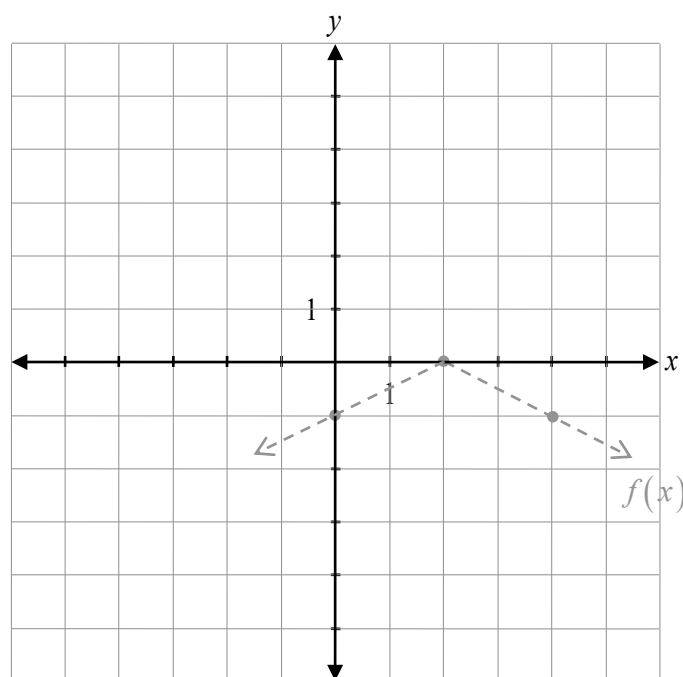
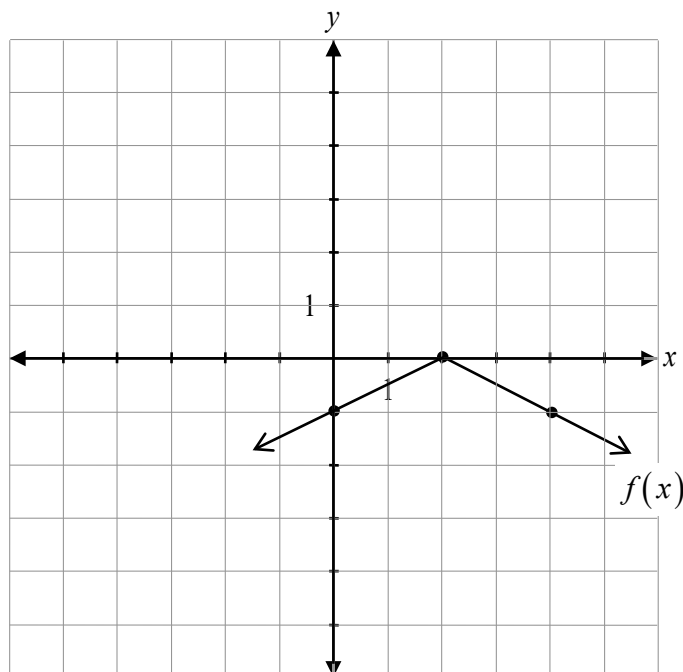
Determine the number of ways that 3 teachers and 9 students can be selected given that Mr. Jones and Mrs. Carol, two of the teachers, must be selected to attend the field trip.

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks 106

Given the graph of $f(x)$, sketch the graph of $y = -f(2x)$.



The graph of $f(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $f(x)$

Question 7

1 mark

107

There are 5 roads between Anneville and Berrybourg, and 2 roads between Berrybourg and Carriton.

Determine how many ways Blake can travel from Anneville to Carriton and back to Anneville, given the following conditions:

- he must travel through Berrybourg in both directions
- he cannot use the same road twice

Determine which term contains x^0 in the binomial expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Question 9

2 marks 109

Given $f(x) = x^3 + 1$, determine the equation of $f^{-1}(x)$.

Question 10

1 mark

110

Guillermo was asked to determine the number of ways to select a president, a vice president, and a treasurer from a group of 11 people.

His solution: ${}_{11}C_3$.

Explain why he should have used a permutation instead of a combination.

Question 11

3 marks 111

Prove the following identity for all permissible values of x .

$$\frac{\csc^2 x \sec x}{\tan x + \cot x} = \csc x$$

Left-Hand Side	Right-Hand Side

Question 12

3 marks 112

Determine the value of x , algebraically.

$$5\log_a 2 - \frac{1}{4}\log_a 16 = \log_a x$$

Tamara must determine the factors of $x^4 - 13x + 2x^3 - 14x^2 + 24$.

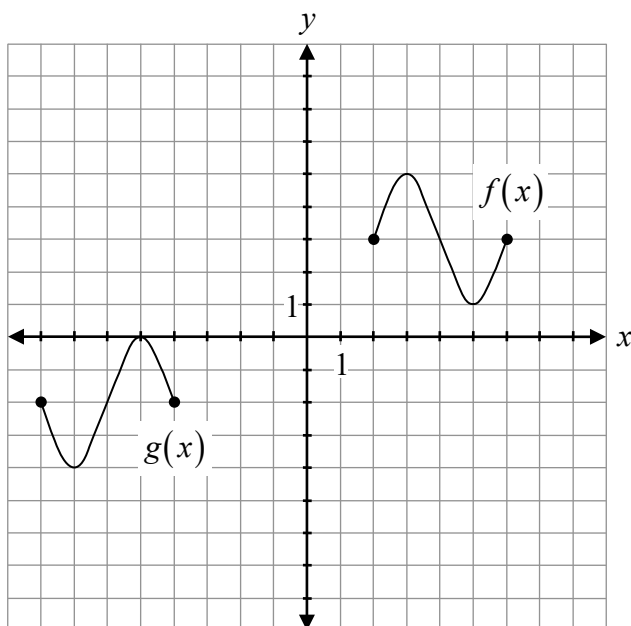
Explain why the coefficients Tamara used to set up her synthetic division are not written correctly.

$$\begin{array}{r|rrrrr} 1 & -13 & 2 & -14 & 24 \end{array}$$

Question 14

3 marks 114

Determine the equation of the graph of $g(x)$ in terms of $f(x)$.



$g(x) =$ _____

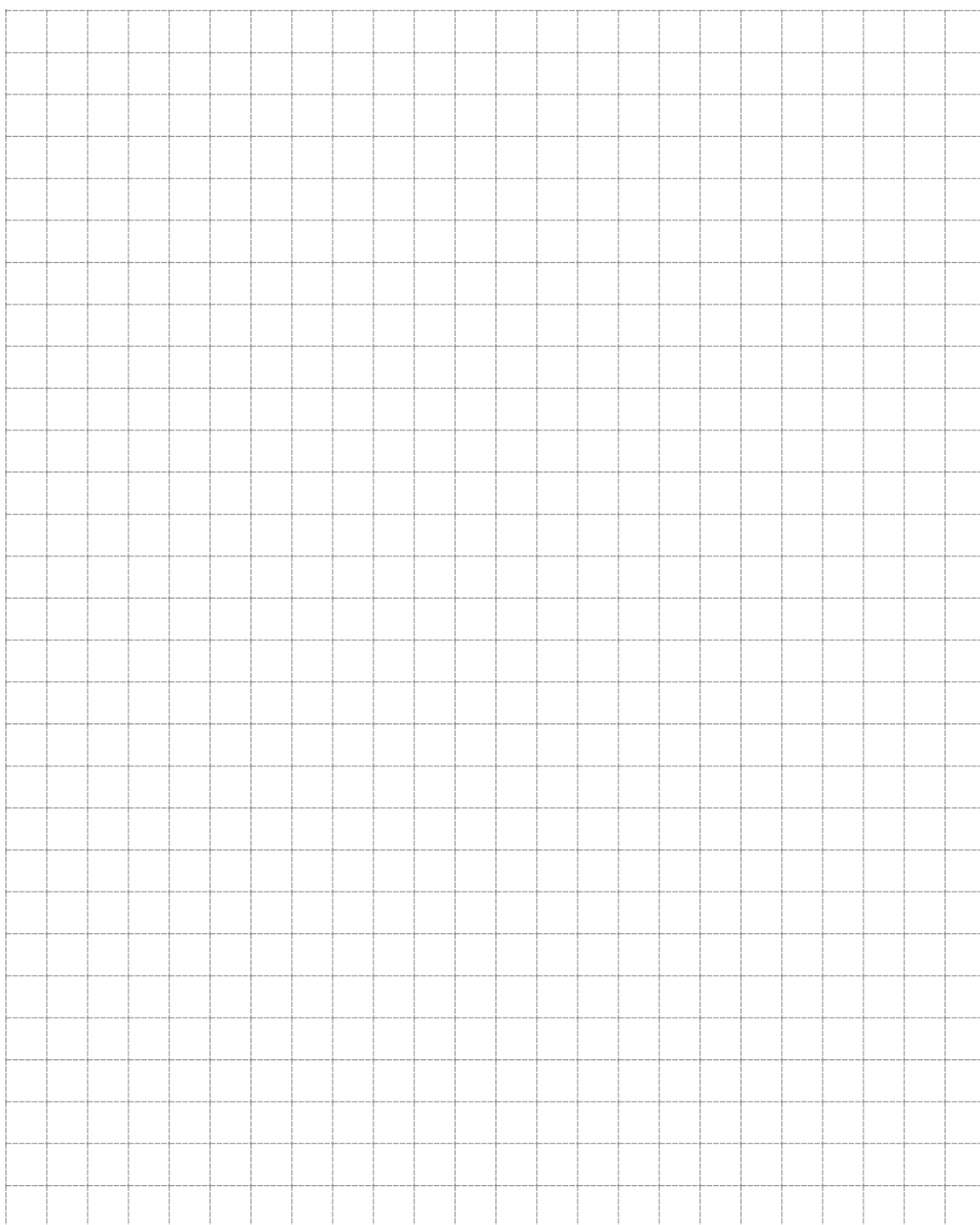
Question 15

2 marks 115

Expand, using the laws of logarithms.

$$\log_2 \left[\frac{(x-1)(x-2)}{x} \right]$$

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.